Introduction to Economics

Class 12 — Perfect competition; monopoly begins

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Universitat de Barcelona · Group A6

2 November 2026

Quiz 9

10 minutes · pens down at 10:10

Circle the letter of your answer in the block at the top.

A question marked + can have more than one correct letter: circle all of them.

Changed your mind? Cross out that circle
and circle the new letter.

Write your name on the block, tear the strip off along the dashed line, hand in *only the strip*.

Plan for today

- A firm that cannot choose its price: how much to produce
- When to produce nothing; the firm's supply curve
- A market's supply: a day of electricity prices
- The long run: entry and exit until economic profit is zero
- Monopoly: a first look

Warm-up: land, labour or capital?

Our print shop's inputs: land, labour or capital?

1. The workshop building it rents
2. The plot of ground under the building
3. The printers' working hours
4. Who owns and supplies all of these?

In pairs, two minutes: four answers; then hands per item.

Factors of production: land, labour, capital (class 3, slide 22).

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   labour: paid wages
4. Who owns and supplies all of these?
   households: as owners and as workers

Factors of production: land, labour, capital (class 3, slide 22).

Warm-up: who should make the summaries?

Exam prep: flashcards and summaries

In one hour — Ana: 6 flashcards or 2 summaries · Ben: 3 flashcards or 1 summary

1. Can both gain by specialising?

2. Ben gets faster: 3 flashcards or 2 summaries. Who makes summaries?

3. After 2: at what price could they swap?

In pairs, one minute: three answers; then hands.

Comparative advantage: the lower opportunity cost (class 3, slide 18).

Warm-up: who should make the summaries?

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1. Can both gain by specialising?
   no: a summary costs each of them 3 flashcards
2. Ben gets faster: 3 flashcards or 2 summaries. Who makes summaries?
   Ben: a summary costs him 1.5 flashcards, Ana 3
3. After 2: at what price could they swap?

Comparative advantage: the lower opportunity cost (class 3, slide 18).

Warm-up: who should make the summaries?

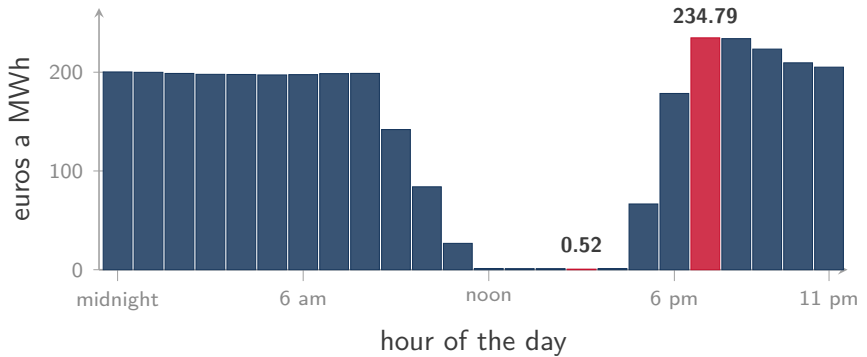
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   between 1.5 and 3 flashcards a summary, e.g. 2

Comparative advantage: the lower opportunity cost (class 3, slide 18).

Sunday 11 October 2026: electricity at 0.52 euros a MWh at 3 pm, 234.79 at 7 pm



Same plants, same market, four hours apart: why?

OMIE (the Spanish–Portuguese wholesale electricity market), day-ahead price, mainland Spain, 11 Oct 2026; each bar averages the hour's four quarter-hour prices. 1 MWh = 1,000 kWh.

Perfect competition: our print shop is a price taker at 4 euros a bag

Tote printing in Barcelona:

- many shops, each one small
- the same printed bag at each
- anyone can open a shop

price taker: no shop can move the price



Competitive market: many sellers of the same good; none can move the price (class 5, slide 8).

Perfect competition: our print shop is a price taker at 4 euros a bag

Tote printing in Barcelona:

- many shops, each one small
- the same printed bag at each
- anyone can open a shop

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Each extra bag adds 4.00 to revenue: **marginal revenue** (MR) = price.

Competitive market: many sellers of the same good; none can move the price (class 5, slide 8).

At 4 euros a bag, profit peaks with four printers: 220 bags, 220 euros a day

printers	bags a day	total cost	MC a bag ΔTC ÷ Δbags
0	0	180	—
1	40	300	3.00
2	120	420	1.50
3	180	540	2.00
4	220	660	3.00
5	240	780	6.00

Total cost (TC) and marginal cost (MC), the cost of one more bag: class 11, slide 10.
Costs here: those paid out.

At 4 euros a bag, profit peaks with four printers: 220 bags, 220 euros a day

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0	0	180	—	0
1	40	300	3.00	160
2	120	420	1.50	480
3	180	540	2.00	720
4	220	660	3.00	880
5	240	780	6.00	960

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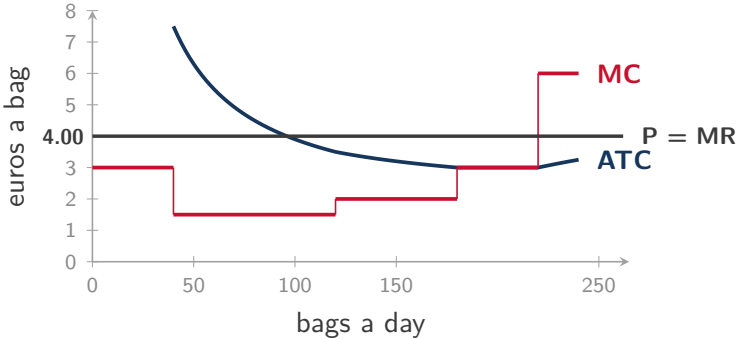
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printers	bags a day	total cost	MC a bag ΔTC ÷ Δbags	revenue 4.00 × bags	profit revenue − cost
0	0	180	—	0	−180
1	40	300	3.00	160	−140
2	120	420	1.50	480	60
3	180	540	2.00	720	180
4	**220**	**660**	**3.00**	**880**	**220**
5	240	780	6.00	960	180

The fifth printer's 20 bags cost 6.00 each and sell for 4.00: profit falls.

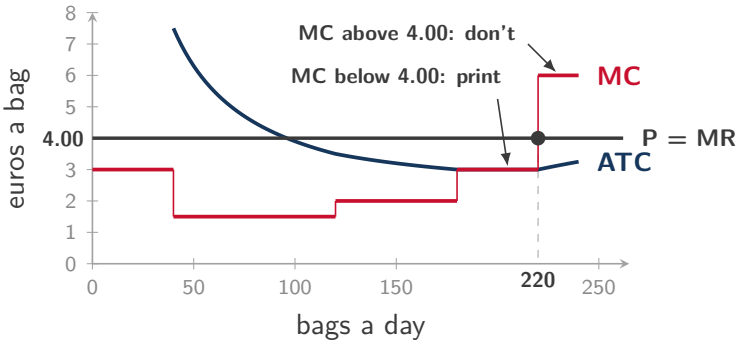
Total cost (TC) and marginal cost (MC), the cost of one more bag: class 11, slide 10.
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Print while marginal cost is below the price: 220 bags, profit 220 a day



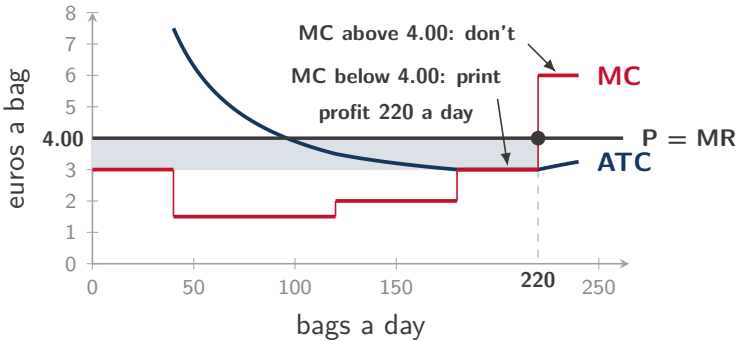
MC: what one more bag adds to total cost · ATC, average total cost: total cost ÷ bags (class 11, slide 10).

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Print while marginal cost is below the price: 220 bags, profit 220 a day



$$\text{Profit} = (\text{price} - \text{ATC}) \times \text{bags} = (4.00 - 3.00) \times 220 = 220 \text{ a day.}$$

MC: what one more bag adds to total cost · ATC, average total cost: total cost ÷ bags (class 11, slide 10).

In pairs, three minutes: a strawberry farm at the market price

A Maresme farm sells strawberries at the market price.
Extra cost of each further 100 boxes a day:

boxes a day	0→100	100→200	200→300	300→400	400→500
euros a box	4	5	7	10	14

1. Price 8 euros a box: how many boxes a day?

2. The price rises to 12: how many now?

3. At 8 euros, what do boxes 301–400 do to profit?

One answer per question; then hands.

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1. Price 8 euros a box: how many boxes a day?
   300 boxes a day: the next 100 cost 10 each
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   300 boxes a day: the next 100 cost 10 each
2. The price rises to 12: how many now?
   400 boxes a day: a higher price, more boxes
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   300 boxes a day: the next 100 cost 10 each
2. The price rises to 12: how many now?
   400 boxes a day: a higher price, more boxes
3. At 8 euros, what do boxes 301–400 do to profit?
   profit falls by 200 euros: $100 \times (8 - 10)$

Below 2 euros a bag the shop prints nothing: the rent is paid either way

price a bag	**4.00**
best if open	220 bags
revenue	880
printers' wages	480
rent and press	180
profit if open	+220
profit if shut down	−180
so	print

AVC, average variable cost: printers' wages ÷ bags, lowest 2.00 · fixed: 180 whatever the output (class 11, slides 9–10) · sunk: already paid, gone whatever you choose (class 2, slide 23).

Below 2 euros a bag the shop prints nothing: the rent is paid either way

price a bag	4.00	2.50
best if open	220 bags	180 bags
revenue	880	450
printers' wages	480	360
rent and press	180	180
profit if open	+220	−90
profit if shut down	−180	−180
so	print	print

AVC, average variable cost: printers' wages ÷ bags, lowest 2.00 · fixed: 180 whatever the output (class 11, slides 9–10) · sunk: already paid, gone whatever you choose (class 2, slide 23).

Below 2 euros a bag the shop prints nothing: the rent is paid either way

price a bag	4.00	2.50	1.80
best if open	220 bags	180 bags	120 bags
revenue	880	450	216
printers' wages	480	360	240
rent and press	180	180	180
profit if open	+220	−90	−204
profit if shut down	−180	−180	−180
so	print	print	**shut down**

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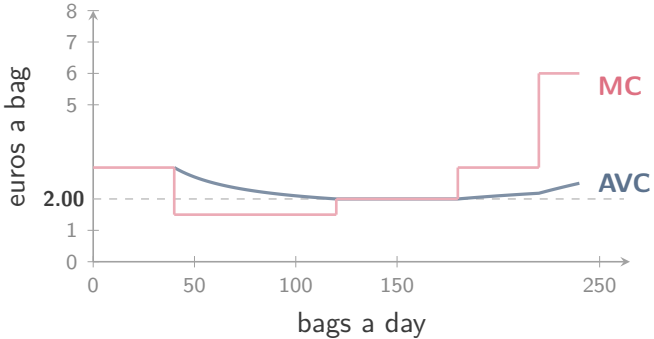
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printers' wages	480	360	240
rent and press	180	180	180
profit if open	+220	−90	−204
profit if shut down	−180	−180	−180
so	print	print	**shut down**

Shut down (for now) if the price is below AVC: here below 2.00.

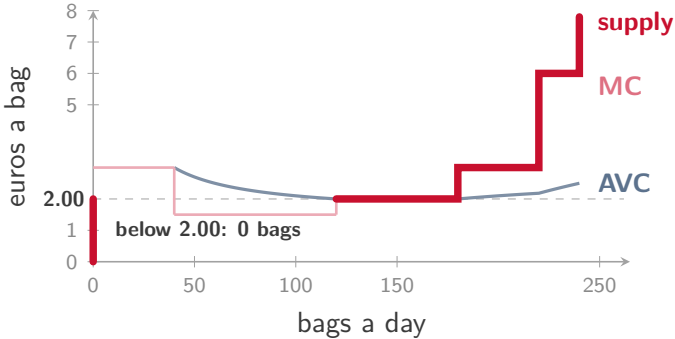
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The shop's supply curve: its rising marginal cost above the lowest AVC



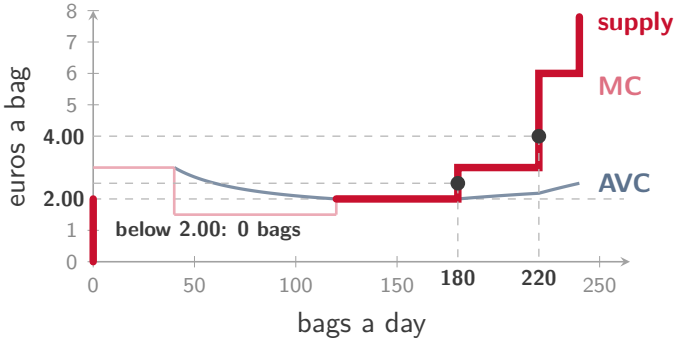
AVC: printers' wages ÷ bags; MC: the cost of one more bag (class 11, slides 10 and 16).

The shop's supply curve: its rising marginal cost above the lowest AVC



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The shop's supply curve: its rising marginal cost above the lowest AVC



A higher price, more bags: supply slopes up because MC rises.

AVC: printers' wages ÷ bags; MC: the cost of one more bag (class 11, slides 10 and 16).

A food truck's festival week: open today? (optional)

Pitch for the week: 600, paid, no refund · each day it opens: ingredients and a helper, 250

1. Sunday's sales would be 300. Open?
2. Rain: Monday's sales would be 200. Open?
3. Does the 600 change either answer?

In pairs, two minutes: three answers.

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   no: opening loses 50 more
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   yes: 300 covers the day's 250
2. Rain: Monday's sales would be 200. Open?
   no: opening loses 50 more
3. Does the 600 change either answer?
   no: the 600 is sunk, paid either way

Break

Back at 11:00

Problem set 2: due Thursday 5 November, on paper

Each plant runs while the price covers its extra cost of a MWh

	Solar farm	Gas plant
extra cost of a MWh	about 0	70
3 pm: price 0.52		
7 pm: price 234.79		
a little sun left		

Panels, turbines, loans: paid either way, so they do not decide.

In pairs, one minute: four answers, one per cell.

70 a MWh: gas plus the CO_2 permit (class 11, slide 14) · slide 10's rule: price against average variable cost; here every MWh costs the same, so that is the extra cost of a MWh.

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	Solar farm	Gas plant
extra cost of a MWh	about 0	70
3 pm: price 0.52	**runs: 0.52 covers its cost**	**off: 0.52 is below 70**
7 pm: price 234.79 a little sun left		

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3 pm: price 0.52	**runs: 0.52 covers its cost**	**off: 0.52 is below 70**
7 pm: price 234.79 a little sun left	**runs: 234.79 covers its cost; little sun left to sell**	

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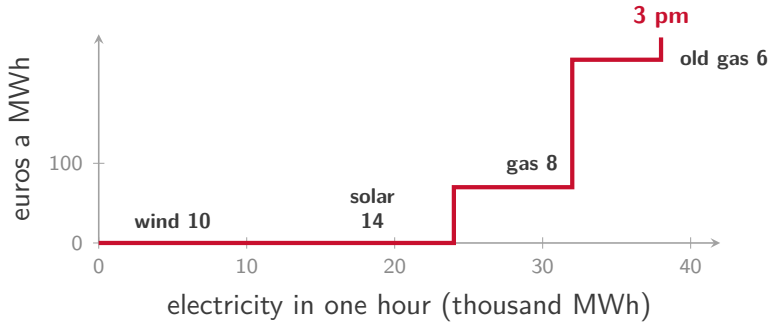
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7 pm: price 234.79 a little sun left	**runs: 234.79 covers its cost; little sun left to sell**	**runs: 234.79 covers 70**

Panels, turbines, loans: paid either way, so they do not decide.

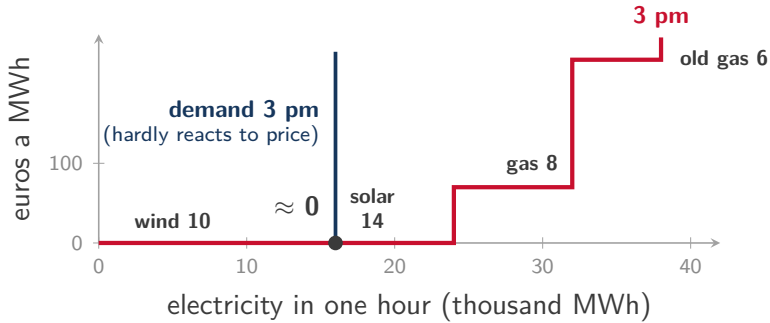
70 a MWh: gas plus the CO_2 permit (class 11, slide 14) · slide 10's rule: price against average variable cost; here every MWh costs the same, so that is the extra cost of a MWh.

Add up the plants: at 3 pm the cheapest are enough; at 7 pm the dearest is needed



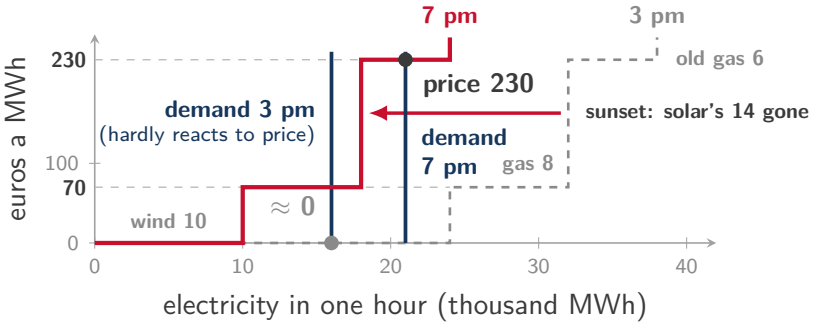
Own round numbers for one hour; real prices on slide 5 (OMIE). Each step is one kind of plant: its width the MWh it can offer, its height its extra cost of a MWh; 'old gas' is an older, dearer plant. Market supply adds the sellers' quantities at each price (class 5, slide 21).

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Market supply: at each price, add the MWh every plant offers.

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In pairs, three minutes: three egg farms, one market

Each farm's cost per dozen stays the same up to its limit.

	cost per dozen	dozen a day (at most)
Farm A	1.00	300
Farm B	1.60	200
Farm C	2.20	100

1. Market supply at 2.00 a dozen?
2. At 2.50?
3. At 1.20?
4. Buyers want 550 dozen a day at any price. The price?

One answer per question; then hands.

In pairs, three minutes: three egg farms, one market

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Farm B	1.60	200
Farm C	2.20	100

1. Market supply at 2.00 a dozen?
   500 dozen: A and B; C stays shut
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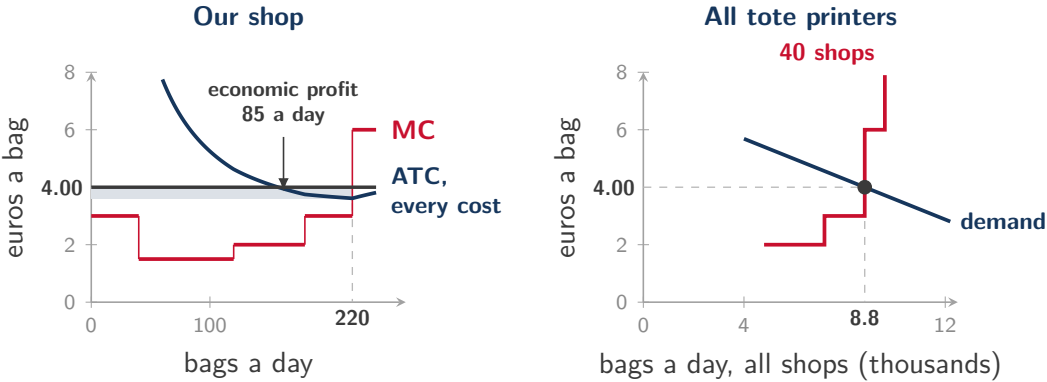
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2. At 2.50?
   600 dozen: all three
3. At 1.20?
   300 dozen: only A
4. Buyers want 550 dozen a day at any price. The price?
   2.20: farm C is needed

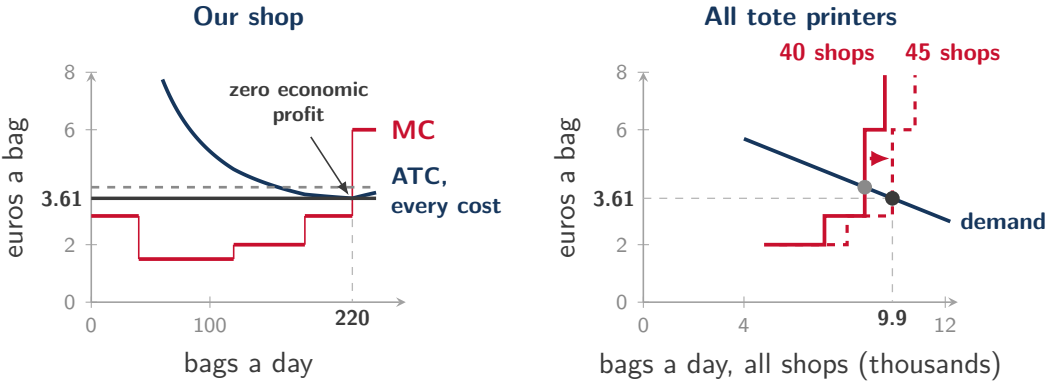
Profit draws new shops: the price falls until economic profit is zero



Each shop: one press · every cost: class 11's costs plus the owner's 135 a day (class 11, slide 6)

long run: time for every cost to change (class 11, slide 18).

Profit draws new shops: the price falls until economic profit is zero



Entry stops when price = lowest ATC (every cost counted): 3.61.

Each shop: one press · every cost: class 11's costs plus the owner's 135 a day (class 11, slide 6)

long run: time for every cost to change (class 11, slide 18).

In the long run the price sits at the lowest ATC: market supply is flat at 3.61



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More demand brings more shops, not a higher price.

Zero economic profit: the owner's accounts still show 2,700 a month — what her job and savings would pay.

In pairs, three minutes: repair stalls in the long run

Phone-screen repair stalls, all alike. Lowest ATC, the owner's time counted: 40 euros a repair. Price today: 55.

1. Economic profit today: positive, zero or negative?
2. Where does the price go?
3. At 40, do the owners quit?
4. A new 5-euro fee on every repair. Long-run price?

One answer per question; then hands.

In pairs, three minutes: repair stalls in the long run

Phone-screen repair stalls, all alike. Lowest ATC, the owner's time counted: 40 euros a repair. Price today: 55.

1. Economic profit today: positive, zero or negative?
   positive: 55 $>$ 40, new stalls open
2. Where does the price go?
3. At 40, do the owners quit?
4. A new 5-euro fee on every repair. Long-run price?

In pairs, three minutes: repair stalls in the long run

Phone-screen repair stalls, all alike. Lowest ATC, the owner's time counted: 40 euros a repair. Price today: 55.

1. Economic profit today: positive, zero or negative?
   positive: 55 $>$ 40, new stalls open
2. Where does the price go?
   down to 40, as new stalls add supply
3. At 40, do the owners quit?
4. A new 5-euro fee on every repair. Long-run price?

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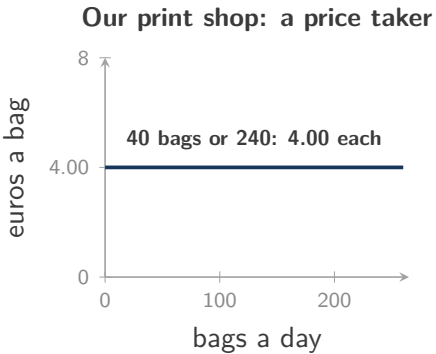
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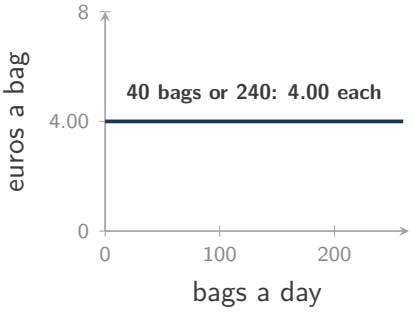
A monopoly is the only seller: to sell more, it must cut its price



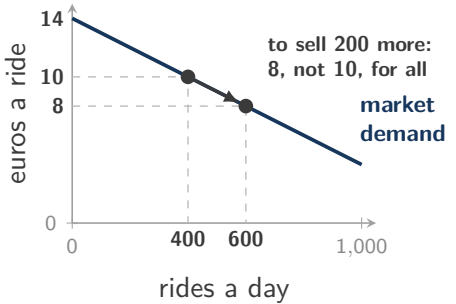
Which point it picks: Monday 9 November (reading Ch. 11).

A monopoly is the only seller: to sell more, it must cut its price

Our print shop: a price taker



A mountain's only cable car: a price maker



A price taker sells any amount at the price; a monopoly picks a point on the demand curve.

Which point it picks: Monday 9 November (reading Ch. 11).

Rivals are kept out by a key resource, a law, or costs that fall with size

A key resource

the only quarry of a rare stone

Economies of scale: ATC falls as output grows (class 11, slide 20).

European Patent Convention, art. 63 (20 years from filing); Ley 24/2013 del Sector Eléctrico, art. 34.2 (Red Eléctrica, transportista único).

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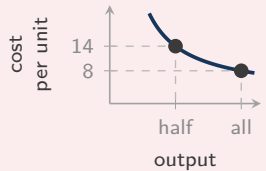
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Falling costs: a natural monopoly

Spain's high-voltage grid: one company, Red Eléctrica



own numbers: one firm, 8 a unit; two firms, half each, 14

What keeps rivals out is a **barrier to entry**.

Economies of scale: ATC falls as output grows (class 11, slide 20).

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Matcha gets popular: which curve moves? (optional)

Barcelona's matcha cafés: many, all alike; lowest ATC 3.50 a cup.

1. A video makes matcha fashionable. Which curve moves?
2. The price rises to 4.50. Does each café's supply curve shift?
3. New cafés open. What moves now?

In pairs, two minutes: three answers.

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3. New cafés open. What moves now?
   market supply shifts right; the price heads back to 3.50

One grid or two: positive or normative? (optional)

1. *"Two grids side by side would cost more per MWh carried."*

2. *"The grid should belong to the state."*

3. A and B both want cheap electricity. A: *"a state company would keep costs lowest."* B: *"a regulated private one would."* Values or a prediction?

In pairs, two minutes: three answers.

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   a prediction: how each would behave

Positive: a claim evidence can test · normative: a claim about what should be (class 4, slide 7).

A price taker produces while marginal cost is below the price.

Below the lowest average variable cost, it produces nothing: fixed costs are sunk.

Market supply adds the firms' quantities; in the long run economic profit is zero.

A monopoly is the only seller: to sell more, it must cut its price.

Before Thursday 5 and Monday 9 November

- Thursday 5 November: poster session, the full two hours; one presenter per group, the whole group answers questions
- Bring your poster printed, A2 portrait; e-mail the PDF by Wednesday 4 November, 8 pm
- Problem set 3 (Block 4): handed out today; due Thursday 12 November, on paper, in class
- Monday 9 November: monopoly; oligopoly and the prisoner's dilemma — read Mankiw & Taylor Ch. 11, pp. 242–261 and Ch. 13, pp. 280–296
- Practise in Sextant: the units the course page lists



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